# Harry Nguyen

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## Education

**Georgia State University** **Expected May 2027**

*Bachelor of Science in Computer Science* · ***GPA: 3.75/4.0***

- **Coursework:** Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- **Honors:** Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

## Work Experience

**TiMoto AI** **Sep 2025 - Present**

*Software Engineer*

- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team — designed, deployed, and operated **distributed production systems** end-to-end
- Designed and maintained **gRPC inter-service layer**; resolved production deadlock under concurrent calls by tracing shared resource acquisition conflicts and redesigning call sequencing — **100% evaluation success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern — **99.9% uptime**, **44% cost reduction** ($40-60/month); auto-rollback on health check failure
- Architected **vLLM inference engine with PagedAttention** for LLM serving — eliminated KV cache memory fragmentation under concurrent load; continuous batching achieving **zero OOM failures** at production traffic

**Google** **May 2025 - Aug 2025**

*Software Engineer Intern — Chrome Browser*

- Architected event-driven **TypeScript/React** system with observer pattern decoupling UI state propagation — **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**
- Designed **C++ IPC transport layer** with Protocol Buffers — selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions
- Collaborated with Chrome infrastructure team to deliver design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

**Develop for Good** **May 2024 - Aug 2024**

*Software Engineer Intern*

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

## Personal Projects

**Pulumi — Contributor | *Go, TypeScript, IaC***

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

## Technical Skills

**Languages**: Python, TypeScript, Go, C++, Java, JavaScript, SQL, Bash, Rust
**Frameworks & Databases**: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB
**Cloud & Infrastructure**: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP
**Distributed Systems**: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture
**ML & AI Infrastructure**: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation
**AI Dev Tools**: Claude Code, GitHub Copilot, Codex, Cursor